MIL-S-13821D(AL)
27 August 1985
SUPERSEDING
MIL-S-13821C(WC)
26 October 1976

MILITARY SPECIFICATION

SCREEN, WELDING, PORTABLE, VINYL COATED FIBERGLASS

This specification is approved for use within the Armament Research and Development Center (Tool and Equipment Engineering Directorate; Standardization and Specification Engineering Division), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

* 1.1 Scope. This specification covers a portable, vinyl coated fiberglass welding screen for use around arc and gas welding operations.

2 APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3431

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MILITARY

MIL-P-116 - Preservation-Packaging, Methods of

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D3951 - Commercial Packaging, Standard Practice for

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORIES INC.

UL 214 - Test for Flame-Propagation of Fabrics and Films

(Applications for copies should be addressed to Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents may also be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3 REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete welding screen for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The welding screen shall be new and of the manufacturer's current model that will meet the requirements specified herein.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the manufacturing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. All parts used in fabrication of the welding screen shall be new. The welding screen shall be complete so that when installed, it can be used for any operation for which it is designed. There shall be no process used for reclaiming any part of the welding screen.

3.4.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching.

* 3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations, however, shall not be employed as repair measures for defective parts.

3.4.3 Surfaces. All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smoothed and all edges shall be rounded or beveled.

3.4.4 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means to prevent loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

* 3.4.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.4.6 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

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3.5 Performance and product characteristics. The welding screen shall embody the following performance and product characteristics.

3.5.1 Functional requirements. The welding screen shall be a portable unit capable of being installed about a welding operation to shield the welding work and prevent the passage of light produced by the electric arc or gas welding outfit.

* 3.5.2 Portability and installation requirements. The construction of the welding screen shall permit installation and set-up by one man without the use of special tools, and fold up for conveyance purposes. The folding or unfolding of the welding screen shall cause no damage to any of its components. It shall be capable of being set-up for use on soil or for shop use on hard surfaces.

* 3.5.3 Size and weight. The welding screen shall embody an effective screening area incorporating a minimum height of 84 inches, commencing at least 3 inches and no more than 12 inches from the ground or floor, and a minimum length of 144 inches. The complete welding screen assembly shall not exceed 65 pounds.

3.6 Components. The welding screen principle parts shall consist of a screen cover assembly and a screen frame assembly.

* 3.6.1 Screen cover assembly. The screen cover assembly shall be made of a one piece green vinyl coated fiberglass in accordance with UL 214 and FED-STD-191 and accordance to size specified in 3.5.3. The cover shall have a minimum thickness of 13 mil (0.013 inch). The cover material shall resist fire, water, mildew, and oils and be washable in accordance with UL 214 and FED-STD-191. The cover shall be attached to the frame's assembly by adjustable ties through heavy duty grommets or as a slip on one piece unit.

* 3.6.2 Screen frame assembly. The frame assembly shall support the vinyl coated fiberglass screen specified in 3.6.1 in compliance with the portability and installation requirements specified in 3.5.2. The frame assembly shall consist of three (3) equal sized panels fabricated of steel or aluminum tubing, solid rod, or formed sheet for the sides, tops and bottoms fastened to corner pieces with set screws. The tubing members shall have an outside diameter of not less than 5/8 inch. The panels shall be hinged together to allow multiple configurations. The hinges shall have the ability to be locked in any position. The legs on the frame shall be of a straight leg design 3 to 12 inches long with capped ends.

* 3.7 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.8 Identification marking. The welding screen shall be marked for identification in accordance with MIL-STD-130 and unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.9 Workmanship. Standards of workmanship shall be such as to assure equipment of a neat appearance, having the stability and efficient operating characteristics as found in commercial equipment of the type specified herein.

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5. Failure of the first article to pass the examination shall be cause for rejection.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of "a" and "b" below. Failure of any item to pass an examination shall be cause for rejection of the item.

a. Product examination (see 4.6).

b. Packaging inspection (see 4.7).

* 4.5 Sampling. Unless otherwise specified (see 6.2.1), sampling for quality conformance inspection shall be in accordance with MIL-STD-105, at the inspection level and Acceptable Quality Level (AQL) as follows:

Inspection	Inspection Level	AQL
Production examination (see 4.6)	II	2.5
Packaging inspection (see 4.7)	S-4	4.0

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* 4.6 Production examination. Visually and manually examine the welding screen to determine conformance with the requirements of 3.3 through 3.9. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the welding screen has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection.

* 4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of section 5.

5 PACKAGING

5.1 Preservation. Preservation shall be level A or industrial, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Components of the welding screen shall be cleaned and dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.2 Preservation. The method of unit protection shall conform to method III requirements of MIL-P-116. The screen shall be collapsed and rolled into a neat bundle. Place the screen into a close-fitting fiberboard box conforming to PPP-B-636, class weather-resistant. The box closure shall be in accordance with the appendix to the box specification.

5.1.2 Industrial. The welding screen shall be commercially preserved in accordance with the requirements of ASTM D3951-82.

5.2 Packing. Packing shall be level A or B, or industrial, as specified (see 6.2.1).

5.2.1 Level A. Welding screens shall be packed in a snug-fitting shipping container conforming to PPP-B-601, overseas type. The gross weight of the shipping container shall be governed by the limitations of the box specification. The box closure shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Welding screens shall be packed as specified in 5.2.1 except the shipping container shall conform to domestic type.

5.2.3 Industrial. Pack welding screens in accordance with the requirements of ASTM D3951-82.

5.3 Marking.

5.3.1 Military. When military levels of protection are specified, marking shall be in accordance with MIL-STD-129 and any special markings required by the contract or purchase order.

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5.3.2 Industrial. When industrial packaging is specified, marking shall be in accordance with ASTM D3951-82 and any special markings required by the contract or purchase order.

6 NOTES

6.1 Intended use. The portable collapsible welding screen covered by this specification is intended for use around arc and gas welding operations.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article, when required (see 3.1).

c. Painting, if different (see 3.7).

d. Identification marking, if different (see 3.8).

e. First article inspection, if different (see 4.3).

f. Quality conformance inspection, if different (see 4.4).

g. Sampling, if different (see 4.5).

h. Specify level A or industrial preservation (see 5.1); specify level A or B, or industrial packing (see 5.2).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-13821	Screen, Welding, Portable, Vinyl Coated Fiberglass

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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